

ERP ↔ Thaisausage Integration Test Plan

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Audience: ERP team, Thaisausage IT/Operations and eVRP test coordinator
System under test: <https://thaisausage.krs.co.th>

1. Test objective

ยืนยันการรับ-ส่งข้อมูลระหว่าง ERP และ thaisausage ตั้งแต่การแจ้งเหตุการณ์ SO การอ่าน SO จาก SQL Server แบบ SELECT-only การส่ง standard JSON ไปยัง eVRP การรับ DO เข้าพื้นที่ staging และการตอบกลับสถานะให้ผู้เรียก โดยไม่เปลี่ยนข้อมูลจริงใน ERP และไม่ส่งข้อมูลจริงไป eVRP จนกว่าจะผ่าน UAT และได้รับอนุมัติ

2. Scope and direction

Direction	Flow	Current scope	Expected side effect
ERP → Thaisausage	direct SO / DO receipt; legacy webhook retained	Yes	SQLite submission/staging only
Thaisausage → ERP	HTTP response to ERP caller	Yes	response only; no ERP write-back
ERP SQL → Thaisausage	approved SO SELECT	Yes	SELECT-only
Thaisausage → eVRP	mapped SO submission	realtime worker/mock first	no real eVRP request in dry-run
eVRP → Thaisausage	response/status	mock/UAT	submission state and reference
Thaisausage → ERP	asynchronous callback	Not available	separate future contract

3. Safety rules

1. Local tests use temporary SQLite and mocked ERP/eVRP connectors. 2. Deployed smoke uses `dry_run=true` and `sync.enabled=false`. 3. Real ERP SQL Server may be queried only for SO numbers explicitly approved by the user. 4. SQL must be the reviewed SELECT query with a read-only account. No INSERT, UPDATE, DELETE, MERGE, EXEC, DDL or ERP status update. 5. DO tests use mock payloads and must prove `erp_write=false`. 6. No real eVRP submission before the UAT window, test token and sign-off are approved. 7. Never log API keys, SQL passwords, full customer data or unredacted payloads.

4. Entry criteria

- ☐ HTTPS, `/health`, `/docs` and `/openapi.json` return successfully.
- ☐ ERP has received the OpenAPI contract and has a test sender.
- ☐ Test API key is configured through secret storage.
- ☐ `dry_run=true` and `sync.enabled=false` are confirmed.
- ☐ Approved SQL query, field mapping and approved SO list are recorded.
- ☐ Test owner, observer and rollback contact are available.
- ☐ eVRP test URL/token and test master data are available before eVRP UAT.
- ☐ Realtime worker is enabled in the test environment; schedule is recovery-only.

5. ERP → Thaisausage test cases

ID	Scenario	Action	Expected result	Evidence
ERP-01	Health/contract	GET /health , /docs , /openapi.json	HTTP 200; no secret exposed	redacted response
ERP-02	Valid SO hook	POST /api/v1/erp/hooks/order-ready with unique event_id	202, state=received , sweep=queued	event_id/status
ERP-03	Hook replay	resend identical body	idempotent; one logical event	hook count/status
ERP-04	Hook conflict	same event_id, changed order data	409 or safe conflict; no second action	response
ERP-05	Invalid hook	missing ID/unknown event type	422; not processed	response
ERP-06	Valid direct SO	POST /api/v1/erp/orders with standard JSON	200/202; request stored	request_id/status
ERP-07	SO replay	same request_id and identical payload	idempotent; no duplicate send	submission lookup
ERP-08	Changed payload	same request_id with changed line	409; operator review	response
ERP-09	Validation	bad date, quantity, hub or duplicate order	422; no outbound attempt	response
ERP-10	DO staging	POST /api/v1/erp/do-received mock receipt	202, state=staged , erp_write=false	receipt/status
ERP-11	DO replay/conflict	same receipt, then changed payload	replay safe; changed payload 409	receipt lookup
ERP-12	Authentication	no/wrong/correct Bearer key	401/401/accepted	status only
ERP-13	Network retry	timeout/disconnect then retry same body	same identity; reconcile review state	IDs/state
ERP-14	Scheduled approved SO	run the schedule with a reviewed query	only approved SOs are selected; no Hook required	timestamps/query result

6. Thaisausage → ERP SQL Server test cases

ID	Scenario	Action	Expected result	Evidence
SQL-01	Connection	use connection string plus ERP_SQL_USER/PASSWORD	connects; credentials not logged	redacted check
SQL-02	Approved SO	run reviewed query for one approved SO	rows read and grouped correctly	row count/order
SQL-03	Unknown SO	query test/nonexistent key	empty/controlled review; no broad sweep	result count
SQL-04	Mutation guard	try mutation/non-SELECT query	rejected before execution	rejection code
SQL-05	Mapping	COD, credit, free item, Thai text, decimals, multi-line	standard JSON matches contract	redacted JSON
SQL-06	Scheduled sweep	run approved-SO schedule in controlled mode	correct submission preview/state; no ERP write	run/result state
SQL-07	SQL failure	simulate timeout/unavailable server	controlled review; scheduler stays healthy	redacted logs
SQL-08	Approved schedule	run configured approved-orders query with IsApprSo = 1	only approved SOs are returned; no Hook required	query and result count

7. Thaisausage → eVRP and response tests

ทดสอบด้วย mock eVRP ก่อนเสมอ และใช้ eVRP UAT เท่านั้นเมื่อได้รับอนุมัติ

ID	Scenario	Action	Expected result	Evidence
VRP-01	Valid outbound	submit mapped mock SO	token header reaches mock only	redacted request
VRP-02	Accepted	mock HTTP 200 with reference	state sent ; reference stored	submission
VRP-03	Temporary failure	mock 502/timeout	needs_review ; no automatic resend; operator reconciliation	redacted response
VRP-04	Ambiguous timeout	accept then close connection	no blind changed-payload resend	review item
VRP-05	Business reject	mock 422	not marked sent; actionable error	response
VRP-06	Duplicate	repeat identical request	idempotent logical result	ID/reference
VRP-07	Auth failure	mock 401/403	no uncontrolled loop; alert operator	status

ERP must treat 202 as accepted for processing, not proof of eVRP completion. ERP may retry transport failures with the same identity. Thaisausage does not run an automatic retry worker or attempt history yet. A 409 stops automation. A 422 requires payload correction. `needs_review` requires operator reconciliation.

8. Test execution record

For every case record: test run ID, environment, Bangkok time, tester, endpoint, HTTP status, request/event/receipt ID, expected state, actual state and redacted evidence path.

At the end record:

- ERP rows before = ____; ERP rows after = ____; mutation check = pass/fail
- Real eVRP send = not used / approved UAT
- Passed = ____; failed = ____; blocked = ____
- Defects, owner, corrective action and retest result

9. Exit gates and sign-off

Gate A — technical smoke

- ☐ ERP-01 to ERP-05 pass.
- ☐ HTTPS, authentication and CSP/Swagger are verified.
- ☐ No secret appears in evidence.

Gate B — controlled data flow

- ☐ ERP-06 to ERP-14 pass with mock/staging data.
- ☐ SQL-01 to SQL-07 pass using only approved SO SELECTs.
- ☐ DO receiver proves `erp_write=false`.
- ☐ ERP row/status counts are unchanged.

Gate C — eVRP UAT

- ☐ VRP-01 to VRP-07 pass against eVRP test environment.
- ☐ eVRP confirms customer, item, hub and remote reference mapping.
- ☐ Retry, duplicate and recovery behavior is accepted.
- ☐ Production token, live mode and schedule are approved separately.

Role	Name	Result	Date/signature
ERP owner		Pass / Fail	
Thaisausage IT		Pass / Fail	
eVRP coordinator		Pass / Fail	
Business/data owner		Pass / Fail	

10. Related contracts

- Swagger: <https://thaisausage.krs.co.th/docs>
- OpenAPI: `docs/openapi.json`
- ERP guide: `docs/ERP_Integration_Guide.md`
- Operations guide: `docs/Thaisausage_Operations_Guide.md`